Where K_b = molal boiling point elevation constant or Ebullioscopic constant.

If m=1, then $\Delta T_b = K_b$;

Hence, K_b is equal to the elevation in boiling point for 1 molal solution. K_b is calculated by the following expression

$$K_b = \frac{RT^2 M_{solvent}}{\Delta H_{vapourisation}}$$

Determination of molar mass of solute from elevation of boiling point

If the solution is prepared by dissolving w_B g of solute in w_A g of solvent, then the molality is,

$$m = \frac{\text{Number of moles of solute} \times 1000}{\text{weight of solvent in grams}} \quad \ldots\ldots\ldots\ldots (9.25)$$

$$\text{Number of moles of solute} = \frac{w_B}{M_B} \quad \ldots\ldots\ldots\ldots (9.26)$$

Where, M_B=molar mass of the solute

Therefore,

$$m = \frac{w_B \times 1000}{M_B \times w_A} \quad \ldots\ldots\ldots\ldots (9.27)$$

and

$$\Delta T_b = \frac{K_b \times w_B \times 1000}{M_B \times w_A} \quad \ldots\ldots\ldots\ldots (9.28)$$

Molar mass can be calculated by using (9.28)

$$M_B = \frac{K_b \times w_B \times 1000}{\Delta T_b \times w_A} \quad \ldots\ldots\ldots\ldots (9.29)$$

Table 9.3 Molal boiling point elevation constant 'K_b' for some solvents

S. No.	Solvent	T_b° (K)	K_b (K kg mol^{-1})
1.	Water	373.15	0.52
2.	Ethanol	351.5	1.20
3.	Benzene	353.3	2.53
4.	Chloroform	334.4	3.63
5.	Ether	307.8	2.02
6.	Carbon tetrachloride	350.0	5.03
7.	Carbon disulphide	319.4	2.42
8.	Acetic acid	391.1	2.93
9.	Cyclohexane	353.74	2.79

Example Problem 4

0.75 g of an unknown substance is dissolved in 200 g solvent. If the elevation of boiling point is 0.15 K and molal elevation constant is 7.5 K Kg mol^{-1} then, calculate the molar mass of unknown substance

$\Delta T_b = K_b\, m$

$= K_b \times W_2 \times 1000 / M_2 \times W_1$

$M_2 = K_b \times W_2 \times 1000 / \Delta T_b \times W_1$

$= 7.5 \times 0.75 \times 1000 / 0.15 \times 200$

$= 187.5$ g mol^{-1}

Evaluate Yourself

11) 2.56 g of Sulphur is dissolved in 100g of carbon disulphide. The solution boils at 319. 692 K . What is the molar mass of Sulphur in solution The boiling point of CS_2 is 319. 450K. Given that K_b for CS_2 = 2.42 K Kg mol^{-1}

Unit 9 - jagan.indd 53 27-01-2021 15:06:29